

Retail Sentiment Intelligence

Trust-Aware Sentiment Analysis of Retail Feedback using LLMs

BITS ZG628T • Dissertation • Post Mid-Semester Presentation

Vishal Singh • ID No. 2020AA05641

M.Tech (Artificial Intelligence & Machine Learning)

Birla Institute of Technology & Science, Pilani (WILP)

Dissertation work at Walmart Global Tech, Bengaluru

Supervisor: Mr. Varunendra Pratap Singh — Principal Software Engineer, Walmart Global Tech

Additional Examiner: Ms. Pradnya Kashikar — BITS Pilani

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Agenda

Focus of today — post mid-semester work only

1 Since Mid-Sem — What's New

Post-midsem workstream map

2 Review & Validate (HITL)

Human-in-the-loop correction workflow

3 Smart Reply — Triple Draft

GPT-4o + Mistral 7B + Smart Composer

4 Smart Reply — Prompt & Few-Shot

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5 Smart Reply — Worked Example

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Macro-F1 0.6272 → 0.7642 (+13.7 pts)

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Hallucination 50% → 0%, extraction 25% → 75%

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15% flagged; 12 of 15 confirmed by annotator

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Lift-and-shift ready

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Dashboard walkthrough (screenshots)

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RO1-RO4 outcomes + roadmap

Since Mid-Sem — What's New

Mid-sem shipped the pipeline; post-midsem made it usable for analysts

Delivered at mid-sem

- 6-layer offline-first pipeline (25 subreddits)
- ModernBERT fine-tuning up to Stage 2 (F1 0.7285)
- Multi-pass vision with 0% hallucination on pilot
- Trust-score design + weighted formula
- Brand-Health + Alert Feed dashboard pages
- Group-routed email / Slack notifications

Added post-midsem (this deck)

- Review & Validate — human-in-the-loop UI + backend
- Smart Reply Composer — dual-draft (rule + FLAN-T5)
- Learning-loop store — corrections & posted replies
- Post Explorer — multi-facet search
- Post Lifecycle — Kanban with 2-step Resolve
- Insights & Competitor Analysis page
- In-app Notification Centre mirroring the digest
- ModernBERT Stage 3 (final) — F1 0.7285 → 0.7642
- Trust-score end-to-end evaluation (n = 200)
- SQLite schema mirroring Cosmos DB partitioning

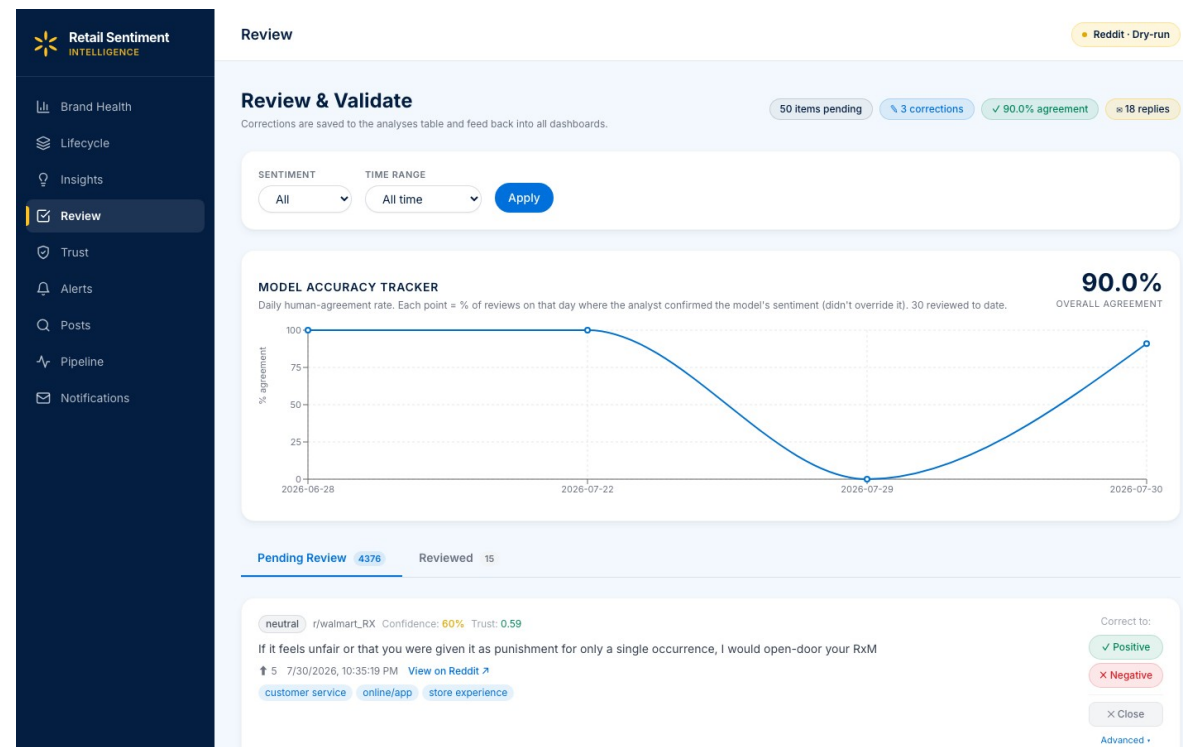
Framing

Mid-sem answered can the pipeline produce trust-weighted sentiment? Post-midsem answers can an analyst act on it, correct it, and feed the corrections back into the model?

Review & Validate — Human-in-the-Loop

Every prediction can be corrected; every correction is logged

- Queue sorted by priority (P1 first) · trust × confidence
- Analyst can override sentiment and aspect tags in one click
- Corrections written to feedback table with analyst + timestamp
- Generate Drafts → two reply options (rule + LLM)
- Analyst picks A or B, edits, and posts to Reddit
- Posted replies also captured → future few-shot pool



Why it matters

Turns a passive dashboard into a training signal — corrections harvested here drive the ModernBERT re-training loop and reply few-shot pool.

Smart Reply Composer — Triple Draft

One prompt, three generators; analyst picks the best draft

Draft A

GPT-4o

Walmart LLM Gateway

Strong reasoning, safest tone
Guardrails via Walmart proxy
~\$0.0002 / reply
Fallback: direct OpenAI

Draft B

Mistral 7B-Instruct

Local Ollama (:11434)

Open-weights, free
Runs fully offline
~15 s warm latency
Strong at retail jargon

Draft C

Smart Composer

Deterministic template

No LLM, always available
Zero latency, zero cost
Content-aware phrase pools
Safety-net fallback

Fallback logic — no draft is ever missing

- GPT-4o unavailable (no gateway key / consumer-ID / network) → Smart Composer draft in slot A
- Mistral unavailable (Ollama not running) → Smart Composer draft in slot B
- Smart Composer always produces slot C — guaranteed reply
- UI labels each card with the actual model used, plus an offline-fallback badge when applicable

Smart Reply — Prompt Design & Few-Shot

Same prompt fed to GPT and Mistral; Smart Composer skips the LLM step

Prompt template (built in `_build_reply_prompt`)

You are a senior Walmart customer-care analyst replying on Reddit.
Write ONE reply to the customer below. Keep it 2-4 sentences,
empathetic, specific to their complaint, no corporate jargon,
no hashtags, no emojis. Do NOT promise refunds you can't verify;
invite them to DM order details if action is needed. Sign off as
a real person, not a brand.

Example customer post: {past_post_1}
Example analyst reply: {past_reply_1}
Example customer post: {past_post_2}
Example analyst reply: {past_reply_2}

Subreddit: r/{subreddit}
Customer ({author}) complaint about: {aspect_1}, {aspect_2}
Customer post:
{post_title + post_body, ≤ 1200 chars}

Reply:

Few-shot source — feedback table

- Top-3 past validated replies for the same aspect
- Written by human analysts, already posted to Reddit
- Refreshed on every Generate click → adapts over time

Style guardrails baked into the prompt

- 2 – 4 sentences, empathetic, specific
- No corporate jargon, no hashtags, no emojis
- No unverifiable refund promises → DM to continue
- Sign off as a person, not a brand

Smart Reply — Worked Example (real post from our benchmark)

id = 1nn7hjxx · r/samsclub · data/benchmark_real_200.jsonl

Customer complaint — r/samsclub · aspects: product_quality, store_experience · trust 0.62 · P2

u/hangry_shopper : "Why do you guys sell whole pizzas made hours ago to customers? The very few times I've gotten a whole pie, it'll be stuff premade and left in the hot case for like an hour 30mins before it's in my hand. How can I tell? They put a sticker with the date and the pizza looks and taste hours old. Pizza is meant to be made to order and waited for."

Draft A · GPT-4o

Really sorry the pizza's been sitting in the hot case that long — that's not the fresh pie experience the club is supposed to deliver. DM me the club number and the day/time of your last visit and I'll flag it to the bakery-café team so the make-to-order flow gets reinforced. — Ravi (Sam's Club Care)

Draft B · Mistral 7B

Totally fair callout — a whole pie that's been in the hot case for 90 minutes isn't the deal. DM me your club number and I'll get the bakery lead to look at their bake-off schedule so the pies you buy are made to order like they should be. — Priya

Draft C · Smart Composer

Hi hangry_shopper, we're really sorry about the quality of what you received on your last visit — a pizza that's been sitting for that long isn't the standard. Please DM us the club number and the day you bought it and we'll get the bakery team on it. — Sam's Club Care

Analyst picks a draft, edits inline, clicks Save & Open Reddit → posted reply is written to feedback (kind = auto_reply_posted) → becomes the next post's top few-shot example. See docs/Sem_4/SMART_REPLY_COMPOSER.md

Learning Loop — Feedback → Retraining

Every human action becomes a training signal — hot loop (live) + warm loop (monthly)

Label / aspect correction	Trust-score override	Posted reply	Lifecycle transition
Review & Validate	Review & Validate	Smart Reply Composer	Kanban board

feedback table (SQLite / Cosmos DB) — one row per human action, JSON payload, partition_key = analyst_id

HOT LOOP • in-context, live

- Every Generate Drafts click queries feedback
- SELECT ... WHERE kind = 'auto_reply_posted'
- ORDER BY created_at DESC LIMIT 5 → top 3 pairs
- Injected into the prompt as few-shot examples
- Effect visible on the next reply — no training

WARM LOOP • supervised, monthly

- Export corrections since last train date
- Append to Walmart-200 → Walmart-N (augmented)
- Rerun ModernBERT Stage-3 with 5-fold OOF CV
- New checkpoint wins only if OOF F1 improves
- Symlink flipped in config/models.yaml

Two failure modes eliminated by design

(1) hot loop uses in-context learning — no retrain, no catastrophic forgetting. (2) warm loop uses 5-fold OOF CV — no leakage between train and eval sets.

Post Explorer — Multi-Facet Search

Find the exact 20 posts an analyst needs to look at today

Facets

- Sentiment · neg / neu / pos
- Confidence slider (0.50 – 1.00)
- Trust-score slider (0.00 – 1.00)
- Subreddit — multi-select from 25 tracked
- Aspect — 8-item retail taxonomy
- Date range — today / week / month / custom
- Full-text search — title + body

Brand Health

Lifecycle

Insights

Review

Trust

Alerts

Posts

Pipeline

Notifications

Posts

Reddit · Dry-run

Post Explorer

Search the analyzed-post archive by subreddit, sentiment, aspect or trust score.

SUBREDDIT

SENTIMENT

ASPECT

RANGE

MIN TRUST

LIMIT

Search

e.g. walmart

All

e.g. pricing

All

0.0-1.0

50

Showing 50 of 83,006 matches (raise the Limit to see more)

neutral

r/WalmartEmployees

↑ 1

7/31/2026, 3:33:40 PM

trust 0.55

onepay

newbie here. how does instapay work on the onepay app?

View on Reddit

negative

r/AmazonPrime

↑ 1

7/31/2026, 3:33:25 PM

trust 0.66

Amazon promised me a \$150 shipping refund in chat to make me return an item, then refused to pay. Can they just lie like that?

Hey everyone, I need some advice or to know if anyone else has dealt with this nightmare. I recently had to return an item to Amazon internationally. When I was setting up the return through live chat, the support agent explicitly told me: "You can use any carrier to return the item. You need to pay to return the item. But the good thing is, we can help you with the refund for return shipping charges up to \$150." Because of that exact, written promise, I went ahead, paid out of pocket, and shipped the...

View on Reddit

negative

r/eCommerce

↑ 1

7/31/2026, 3:27:20 PM

trust 0.66

Struggling with cancelled parcels in fulfillment

We've been seeing a rise in cancelled orders on TikTok, Lazada, and Shopee that are already processed and packed in our fulfillment area. The problem is, by the time we realize they're cancelled, the items are fully packed— meaning wasted packaging materials and extra work. I'm trying to figure out how to catch these cancellations earlier in the workflow so we can save on cartons, fillers, and waybills. Has anyone here dealt with this? How do you identify cancelled items before they even reach...

View on Reddit

positive

r/CustomerService

↑ 1

7/31/2026, 3:24:38 PM

trust 0.66

Dear ppl who call in to customer service

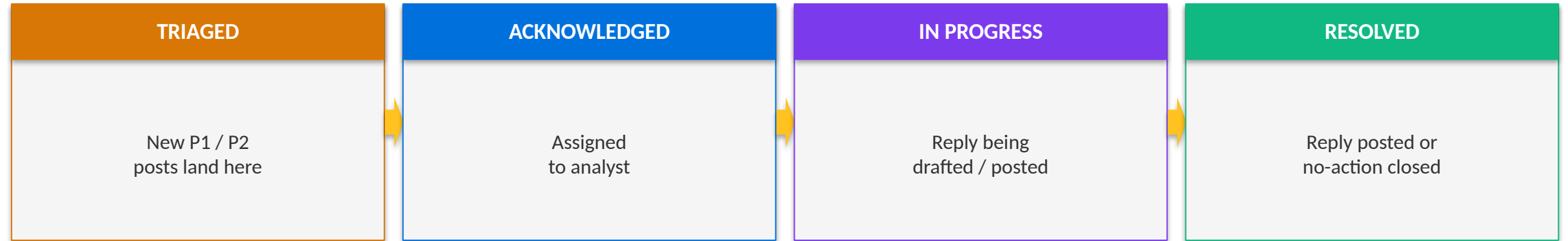
I am a call center agent, I actually enjoy my job! but for those who call into anywhere yelling and fussing about goofy shyt. or a package coming a day late, or ur broke ass begging for a discount... get it together I hope you get hung up on until you know how to talk to people. I hang up on customers all the time soon as

Per-post card

Title + body excerpt · sentiment badge with confidence % · trust tier (H / M / L) · aspect chips · subreddit + timestamp · actions: Review · Add to Lifecycle · View Details.

Post Lifecycle — Kanban Workflow

Track a complaint from Triage to Resolved with SLA visibility



The screenshot shows the 'Post Lifecycle' dashboard in the Retail Sentiment Intelligence application. The dashboard is organized into three main columns: 'Ack & Reply Sent', 'Actionable Items', and 'Resolved'. Each column contains a list of posts with details such as severity (MEDIUM, LOW), source (e.g., r/OGPBackroom, r/WalmartSparkDrivers), and a 'Resolve' button. A sidebar on the left provides navigation options like 'Brand Health', 'Lifecycle', 'Insights', 'Review', 'Trust', 'Alerts', 'Posts', 'Pipeline', and 'Notifications'. A 'Refresh' button is located in the top right corner of the dashboard area.

Resolve modal — 2-step flow

- Step 1 · save action note + optional LLM reply
- (a) Save & open Reddit → reply copied to clipboard
- (b) OR Resolve (no-reply) → close without posting
- Step 2 · paste on Reddit → Mark Resolved
- All transitions timestamped → SLA analytics

Insights & Competitor Analysis

Strategic view — issue rankings, cross-brand pulse, LLM summaries

Priority-Negatives

Top issues ranked by
volume × severity × recency
by aspect

Competitor Pulse

Walmart vs Costco,
Target, Amazon on shared aspects

Weekly LLM Summaries

Natural-language weekly
digest + action items
+ emerging topics

Aspect Drilldown

8-aspect taxonomy
per-aspect sentiment trend
+ representative posts

Screenshot on Slide 17 (Live Demo grid) — insights_competitor page.

Notification Centre

In-app mirror of the group-routed email / Slack digest

- Every P1 / P2 email or Slack alert is mirrored in the app
- Notifications grouped by subreddit ↔ business team
- Read / unread state persisted per analyst
- Deep-links open the post directly in Review & Validate
- Powered by the same alerts rows that drive email/Slack



- Brand Health
- Lifecycle
- Insights
- Review
- Trust
- Alerts
- Posts
- Pipeline
- Notifications**

Notifications

Reddit - Dry-run

Notification Configuration

Configure which subreddit groups receive alerts for P1/P2 negative posts.

SENDER EMAIL

vishal.singh1@walmart.com

DriverRelatedPost

3 subreddits · 0 recipients · retailSentimentIntelligence

r/Sparkdriver

r/walmartgo

r/WalmartSparkDrivers

P1 + P2

GIF_

1 subreddit · 0 recipients · retailSentimentIntelligence

r/OGPBackroom

P1 + P2

SAMSClub

2 subreddits · 0 recipients · retailSentimentIntelligence

r/samsclub

r/Sams_Club

P1 + P2

W+ Membership

1 subreddit · 1 recipient · retailSentimentIntelligence

r/WalmartPlus

P1 + P2

Recent Notification Log

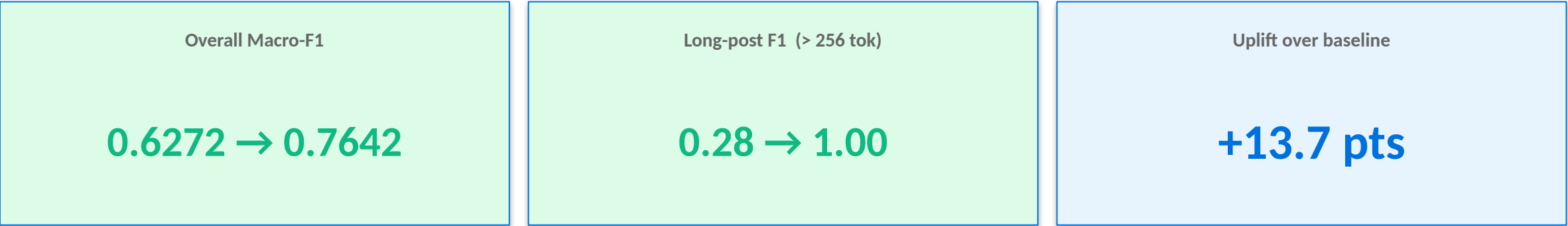
TIME	GROUP	CHANNEL	POST	STATUS
Jul 31, 04:15 PM	DriverRelatedPost	slack	reddit_1v5jl7	FAILED
Jul 31, 04:15 PM	DriverRelatedPost	slack	reddit_1vblyrt	FAILED
Jul 31, 04:15 PM	DriverRelatedPost	slack	reddit_c_p8qqv16	FAILED

Priority thresholds

P1 — trust ≥ 0.70 and confidence ≥ 0.80
P2 — trust ≥ 0.50 and confidence ≥ 0.60
Source: config/models.yaml + dispatcher.py

ModernBERT — Final Training Results

3-stage curriculum, 5-fold out-of-fold CV, n = 200



3-stage curriculum

Stage	Data	Macro-F1	Δ vs baseline
Baseline (Twitter-RoBERTa)	off-the-shelf	0.6272	—
Stage 1	TweetEval sentiment (~45 k tweets)	0.6810	+5.4
Stage 2	GoEmotions-3class Reddit (~54 k, pseudo)	0.7285	+10.1
Stage 3 (final)	Walmart-200 hand-labelled	0.7642	+13.7

Per-length-bucket F1

Bucket	Baseline	ModernBERT	Δ
Short (< 64 tokens)	0.7500	0.7800	+3.0
Medium (64 - 256 tokens)	0.6500	0.7400	+9.0
Long (> 256 tokens)	0.2778	1.0000	+72.2

Vision Pipeline — Multi-Pass Payoff

Same model (Gemma 3 4B), same hardware, smarter calling strategy



Evaluation on 32 retail screenshots

Metric	Single-pass	Multi-pass	Change
Hallucination rate	50 %	0 %	eliminated
Text-extraction success	25 %	75 %	3× better
Retail-signal recall	40 %	81 %	2× better
Fabricated claims	many	0	eliminated
Latency / image (warm)	~5 s	~15 s	3× (accepted)

Trust-Score — End-to-End Evaluation

On the 200-post gold set, does the trust filter catch bad posts?

Low-trust share ($T < 0.4$)

15 %

Human-agreed (annotator cross-check)

12 / 15

Agreement rate

80 %

Weighted formula (decomposition shown inline in dashboard)

$$\text{trust_score} = 0.4 \times \text{metadata} + 0.3 \times \text{dedup} + 0.3 \times \text{llm_credibility}$$

Design principle — flag, don't drop

Low-trust posts are surfaced with an explanatory chip ("promotional / duplicate / thin-metadata") and can be overridden by an analyst in one click — the override is captured for the learning loop. Every constant in the formula has a stakeholder-arguable English rationale in config/models.yaml.

Storage — SQLite → Cosmos DB Lift-and-Shift

Same schema, same partition keys, no calling-code change

Container	Partition Key	Key Fields	Purpose
raw_posts	/subreddit	id, title, body, author_hash, created_utc	Ingested (privacy-safe)
analyses	/subreddit	post_id, sentiment, confidence, aspects, trust	AI analysis results
aggregates	/time_window	subreddit, window, metrics_json	Pre-computed KPIs
alerts	/severity	type, aspect, threshold_breached	Triggered anomalies
feedback	/analyst_id	post_id, correction, reply_text	HITL corrections + replies
notification_log	/group_id	post_id, channel, status, sent_at	Delivery audit trail

Design choices

- SQLite in WAL mode for hot dev data
- data JSON column mirrors Cosmos doc
- Pluggable StorageBackend interface
- Nightly backup + JSONL cost ledger

Migration story

- Swap SQLiteBackend → CosmosBackend
- Same CREATE TABLE mapped to container spec
- Partition keys already match production
- Zero change to pipeline / dashboard code

Evaluation Summary — Post-Midsem Numbers

All results reported against the same 200-post retail-Reddit gold set

Area	Metric	Value	Notes
Sentiment	Macro-F1 (final)	0.7642	+13.7 pts vs RoBERTa baseline
Sentiment	Long-post F1 (> 256 tokens)	1.0000	From 0.2778 baseline
Sentiment	Cross-validation	5-fold OOF	No test-set leakage
Vision	Hallucination rate	0 %	From 50 % on single-pass
Vision	Text extraction success	75 %	From 25 % on single-pass
Vision	Retail-signal recall	81 %	From 40 % on single-pass
Trust	Low-trust share (T < 0.4)	15 %	Of the 200-post set
Trust	Annotator agreement on low-trust	12 / 15	80 %
Ops	Subreddits tracked	25	Six community groups
Ops	Scheduler cadence	6 h + manual	asyncio lifespan task
Ops	Dashboard pages shipped	8	Brand Health · Alerts · Explorer · Review · Lifecycle · Insights · Pipeline · Notifications

Principal Contributions

What this dissertation delivered — end to end

C1

Offline-first RSI pipeline

Ingestion → trust → sentiment → aspects → vision → aggregation → alerts → dashboard. Zero API cost, modular, deployable to Azure.

C2

Fine-tuned ModernBERT (3-stage curriculum)

Macro-F1 0.6272 → 0.7642 overall; 0.28 → 1.00 on long posts. 5-fold OOF cross-validation, offline training.

C3

Multi-pass vision technique on Gemma 3 4B

Hallucination 50 % → 0 %, extraction 25 % → 75 %. Adapts techniques from 5 recent VLM papers without vendor-blocked models.

C4

Interpretable trust score + admission gate

trust × confidence gate flags rather than drops low-credibility posts. Every constant traceable to English rationale in config.

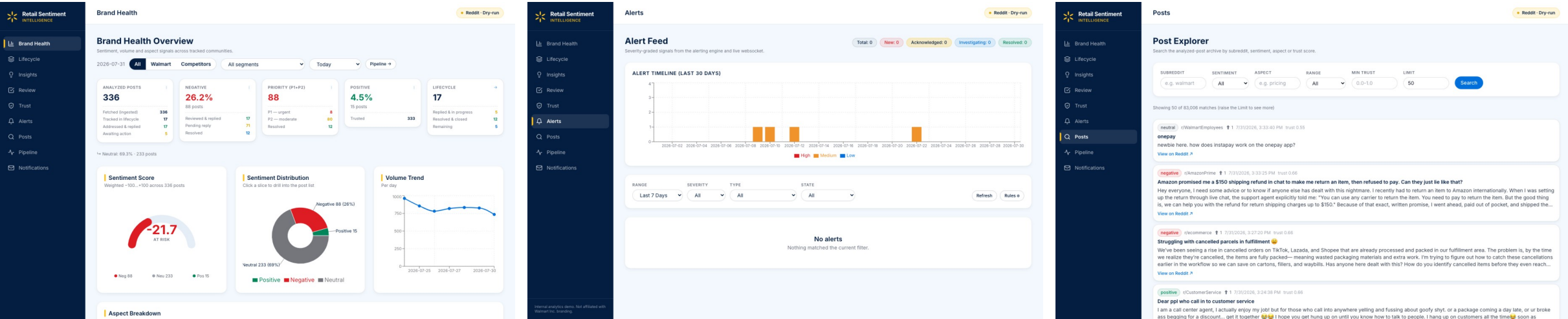
C5

HITL learning-loop dashboard

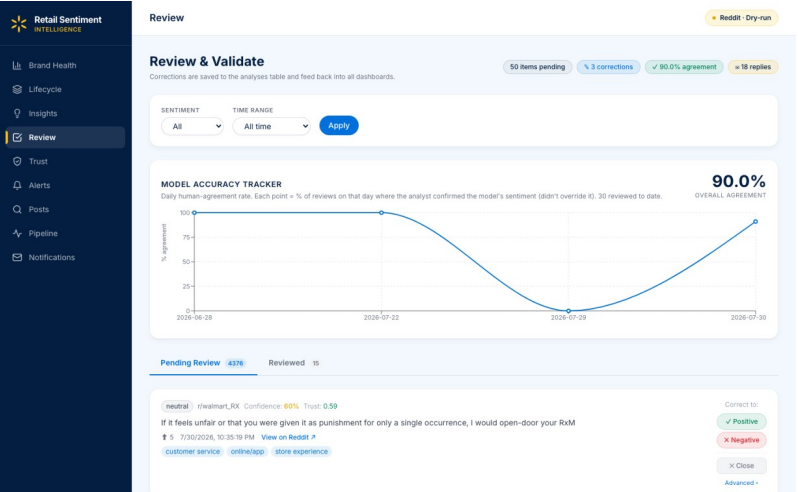
Review & Validate, Lifecycle Kanban, Insights, Notification centre. Corrections & posted replies feed few-shot reply generation and future retraining.

Live Demo — Dashboard Walkthrough

One post → alert → review → reply → lifecycle → resolved

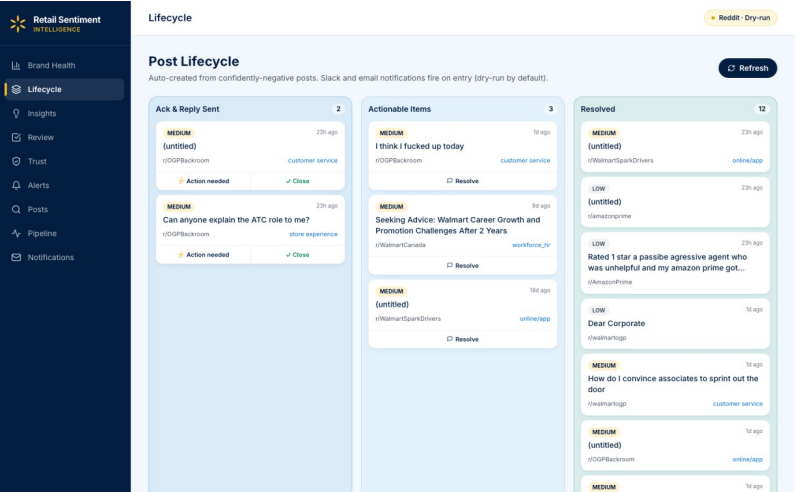


Brand Health



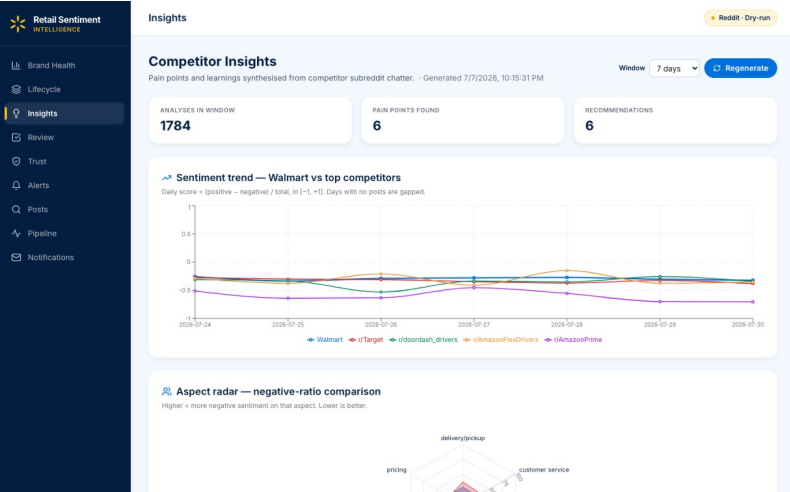
Review & Validate

Alert Feed



Lifecycle Kanban

Post Explorer



Insights

Conclusions — Research Questions Answered

RQ1

Can a fine-tuned encoder beat baselines on Reddit retail sentiment?

YES — ModernBERT 3-stage curriculum raised Macro-F1 from 0.6272 → 0.7642 overall and from 0.2778 → 1.0000 on long posts.

RQ2

Can a compliant open-weights VLM extract structured retail signal from screenshots?

YES — Multi-pass pipeline on Gemma 3 4B reduced hallucination from 50 % → 0 % and lifted text extraction from 25 % → 75 %.

RQ3

Can a defensible trust score filter low-credibility posts without silent drops?

YES — Interpretable 3-part score flagged 15 % of posts; 12 of 15 confirmed by human annotator. All low-trust posts remain visible and analyst-overridable.

RQ4

Can a HITL workflow produce a re-training signal from analyst corrections?

YES — Every correction and posted reply is logged in the `feedback` table → feeds ModernBERT Stage-3 augmentation and FLAN-T5 few-shot pool.

Recommendations — Immediate Follow-Ons

Monthly retraining cadence

Use the HITL feedback store as the incremental labelling stream and re-run the 3-stage curriculum monthly.

Bilingual pass (Hindi + English)

Validate the retail taxonomy with native-speaker analysts before extending to Indian retail communities.

Per-team SLA dashboards

Extend the Post Lifecycle SLA analytics into per-team dashboards once analyst volume passes ~50 posts / day.

Future Work — Model + Product

Directions grouped by layer of the stack

Model-level

- Joint sentiment + aspect head on a shared encoder (~30 % inference saving)
- Distil ModernBERT to ~50 M-parameter student → CPU-only edge inference
- Reasoning-augmented VLM captioning (LLaVA-Next 1.6 B or SmolVLM)
- 3-seed ensemble for tighter F1 variance
- Blind 25-post recheck for defensibility

Product-level

- Auto-reply confidence gate — promote LLM drafts to "queued to send" when analyst edit-distance < threshold
- Predictive P1 forecast — seasonal decomposition on daily counts (24–48 h ahead)
- Bilingual (Hindi + English) taxonomy
- Slack-bot inline notification responses
- Automated retraining pipeline hook

Future Work — Operational

Path from local prototype to Walmart-production service

Kubernetes CronJob

Deploy the pipeline as a scheduled Kubernetes CronJob backed by a managed Postgres instance for multi-analyst concurrency.

Azure Cosmos DB migration

Swap SQLiteBackend for CosmosBackend (schema already mirrors partition design) — no calling-code change.

Walmart ticketing integration

P1 alerts open cases directly in Walmart's internal ticketing so the analyst never leaves the workflow.

Broader ingestion

Extend beyond Reddit to Twitter / X, YouTube comments, and app-store reviews on the same trust + sentiment stack.

Source Code & Deliverables

Repository

https://gecgithub01.walmart.com/v0s01jh/Retail_Sentiment_Intelligence

Deliverable	Path in repo	Contents
Final Report (PDF)	docs/Sem_4/final/FINAL_REPORT_VishalSingh_2020AA05641.pdf	10 chapters + appendices + refs
Final Report (LaTeX)	docs/Sem_4/final/latex/	Reproducible xelatex source
Pipeline core	src/pipeline.py, src/ingestion/, src/analysis/	6-layer async pipeline
Sentiment training	scripts/train_modernbert_sentiment.py	3-stage curriculum runner
Evaluation notebook	evaluation/trust_score_walmart200.ipynb	Reproducible 5-fold OOF eval
Dashboard	frontend/ (React + Vite + Tailwind)	8 pages, live via WebSocket
Reproduction	Appendix B of the report · start.sh	Clone → conda → start

Thank You

Retail Sentiment Intelligence · Post Mid-Semester Presentation

Questions ?

[Repository](#)

[gecgithub01.walmart.com/v0s01jh/Retail_Sentiment_Intelligence](https://github.com/v0s01jh/Retail_Sentiment_Intelligence)

Report: docs/Sem_4/final/FINAL_REPORT_VishalSingh_2020AA05641.pdf

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